Highlights

Knowledge-guided Dual-Space Evolutionary Learning for High-Dimensional Bioinformatics Feature Selection

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- Propose a bi-objective-regulated dual-space evolutionary learning framework (KDS-EL) for high-dimensional bioinformatics feature selection.
- Design a Diversity-regulated Selection (DRS) operator that elevates NHD into a first-class Pareto selection objective to prevent structural collapse.
- Develop a Knowledge-guided Elite Refinement (KER) mechanism that accumulates improvement evidence online to direct targeted bit-flip perturbations.
- Demonstrate synergistic gain of DRS and KER on 7 of 9 benchmark datasets.
- Achieve superior classification accuracy and maximum feature reduction against eight competitors including PSO-CSM (2025) and IGPSO (2024).

Knowledge-guided Dual-Space Evolutionary Learning for High-Dimensional Bioinformatics Feature Selection

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ABSTRACT

Feature selection is essential for reducing dimensionality and improving classifier performance in high-dimensional bioinformatics data, where the exponential search space makes evolutionary methods highly susceptible to structural population collapse and knowledge waste. To address these coupled challenges, we propose **KDS-EL** (*Knowledge-guided Dual-Space Evolutionary Learning*), a novel bi-objective-regulated evolutionary framework that simultaneously optimises a dual-population PSO backbone across two complementary search spaces. First, we design a **Diversity-regulated Selection** (DRS) operator that elevates Normalised Hamming Distance (NHD) from a post-hoc diagnostic metric to a first-class Pareto selection objective, providing a provable structural diversity lower bound and preventing premature population convergence. Second, we develop a **Knowledge-guided Elite Refinement** (KER) mechanism that accumulates improvement-event evidence from the evolutionary trajectory into a feature-importance weight vector, and uses this online-learned knowledge to direct targeted bit-flip perturbations of elite solutions without additional fitness evaluations. The two mechanisms are synergistically coupled: DRS's structural diversity ensures heterogeneous improvement signals for KER, while KER's directed perturbations enrich the DRS selection pool. Extensive experiments on nine high-dimensional bioinformatics datasets ($d \in [927, 26{,}877]$) against eight state-of-the-art competitors demonstrate that KDS-EL achieves the highest average Friedman rank of 1.222 (Friedman $\chi^2 = 47.38$, $p = 1.30 \times 10^{-7}$), reduces selected feature counts by up to 88% relative to the base framework, and outperforms recent methods including PSO-CSM (2025) and IGPSO (2024) across all nine datasets with statistical significance. Ablation analysis confirms positive synergistic gain on 7 of 9 datasets, validating the Dual-Space design rationale.

1. Introduction

With the rapid advances in high-throughput sequencing technologies, gene expression and other omics datasets routinely comprise tens of thousands of features measured over fewer than 200 samples [1, 2]. This extreme $d \gg n$ regime—where the number of features d exceeds the number of samples n by orders of magnitude—poses severe challenges for classification algorithms, including the curse of dimensionality, overfitting, and prohibitive computational cost. Feature selection (FS), which identifies a compact and informative subset of the original features, has emerged as an indispensable preprocessing step in bioinformatics pipelines [3, 4]. Among FS paradigms, wrapper-based evolutionary methods that employ metaheuristic search—particularly Particle Swarm Optimisation (PSO)—have demonstrated strong empirical performance owing to their ability to explore the combinatorial binary search space without requiring differentiable objectives [5, 6].

Despite these advances, existing evolutionary FS methods suffer from two fundamental and coupled limitations. *First*, most methods treat classification accuracy (or its surrogate fitness) as the sole selection criterion [7–9], leaving the *structural geometry* of the binary solution space unexploited. In the high-dimensional binary landscape, random feature subsets exhibit low Normalised Hamming Distance (NHD) to one another; when combined with accuracy-only selection, this structural homogeneity drives rapid population collapse into a small region of the search space, prematurely eliminating potentially valuable feature combinations. To date, NHD has been used only as a post-hoc diversity diagnostic, never as a first-class selection objective within wrapper-based evolutionary FS. *Second*, the cumulative evidence of which feature transitions proved beneficial during the search—the *trajectory knowledge*—is discarded at every generation. Each time the personal best (P_{best}) is updated, the information about which features contributed to the improvement is lost, wasting a valuable signal that could guide future search.

To address these coupled challenges, we propose **KDS-EL** (*Knowledge-guided Dual-Space Evolutionary Learning*), a bi-objective-regulated evolutionary framework that operates simultaneously in two complementary search spaces: (i) a *Structural Space*, where a Diversity-regulated Selection (DRS) operator elevates NHD to a first-class Pareto selection objective; and (ii) a *Fitness Space*, where a Knowledge-guided Elite Refinement (KER) mechanism accumulates improvement evidence from the evolutionary trajectory to direct targeted bit-flip perturbations of elite solutions. The two mechanisms are synergistically coupled: DRS maintains structural diversity in the population, ensuring that KER receives heterogeneous improvement signals; conversely, KER's directed perturbations generate high-quality diverse candidates that enrich the DRS selection pool. This synergy is the central design principle of KDS-EL and is empirically validated on 7 of 9 benchmark

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datasets.

The main contributions of this paper are as follows:

(i) We propose KDS-EL, a dual-space evolutionary learning framework for high-dimensional bioinformatics feature selection that simultaneously optimises fitness and structural diversity.

(ii) We design the DRS operator that incorporates NHD as a first-class Pareto selection objective, with a provable diversity lower bound and an adaptive schedule $\alpha_{\text{cd}}(t) = (t/T)^2$ that transitions smoothly from diversity exploration to accuracy exploitation.

(iii) We develop the KER mechanism that accumulates improvement-event evidence into a feature-importance weight vector online, enabling directed elite refinement with no additional fitness evaluations and a single interpretable parameter.

(iv) Extensive experiments on nine bioinformatics datasets against eight competitors—including PSO-CSM (2025) and IGPSO (2024)—demonstrate that KDS-EL achieves the highest average Friedman rank of 1.222, with statistical significance confirmed by Wilcoxon signed-rank tests.

The remainder of this paper is organised as follows. Section 2 reviews related work on evolutionary feature selection, population diversity maintenance, and knowledge-guided methods. Section 3 presents the proposed KDS-EL framework in detail, including the DRS and KER mechanisms with their theoretical properties. Section 4 reports the experimental setup, results, and discussion. Section 5 concludes the paper and outlines future research directions.

2. Related Work

2.1. Evolutionary Feature Selection

Feature selection methods can be broadly categorised into filter, wrapper, and embedded approaches [10–13]. Wrapper methods evaluate candidate feature subsets by training and testing a classifier (e.g., SVM [14] or Random Forest [15]), making them more accurate but computationally more expensive than filter methods. Evolutionary computation (EC) techniques, such as PSO, Grey Wolf Optimizer [16], and Differential Evolution, have been widely adopted as wrapper-based FS solvers due to their ability to navigate the 2^d combinatorial search space without requiring gradient information [17–21]. Binary PSO (BPSO) [22–24] was among the first to adapt PSO for discrete optimisation, encoding feature subsets as binary vectors and using sigmoid transfer functions for position updates. Subsequent variants have improved upon BPSO in various directions. Adaptive inertia weight strategies have been proposed to balance exploration and exploitation across the search trajectory [8, 25–27]. FWPSO [9] introduces a feature-weight-based initialisation mechanism that accelerates convergence by biasing the initial population toward promising regions. SaWDE [28] combines self-adaptive differential selection with PSO, achieving competitive performance with reduced parameter tuning. MEL [29] proposes a dual-subpopulation architecture where one subpopulation performs standard PSO exploration while the other employs knowledge-guided sampling, demonstrating strong results on bioinformatics benchmarks. More recently, PSO-CSM (2025) introduces a comprehensive scoring metric that integrates multiple criteria, and IGPSO (2024) pre-trains a multi-layer perceptron (MLP) to estimate feature importance prior to the search. Despite these advances, the majority of existing methods treat classification accuracy as the sole selection objective, neglecting the structural information embedded in the binary solution space. Table 1 summarises the key characteristics of recent PSO-based FS methods.

2.2. Population Diversity in Evolutionary Feature Selection

Population diversity is widely recognised as a critical factor affecting the performance of evolutionary algorithms, as insufficient diversity leads to premature convergence while excessive diversity slow convergence [32, 33]. In the context of binary feature selection, diversity can be quantified by the Normalised Hamming Distance (NHD), which measures the proportion of differing feature inclusions between two solutions. Several diversity maintenance strategies have been proposed in the evolutionary FS literature. Multi-population approaches [29] split the swarm into subpopulations that explore different regions of the search space. Population reinitialisation methods periodically replace converged individuals with random solutions to restore diversity [33]. Crowding and sharing techniques, originally proposed for multi-objective optimisation [34–36], have been adapted to maintain diversity in FS populations. However, these approaches treat diversity as a secondary concern—a means to avoid stagnation rather than a formal optimisation target. To the best of our knowledge, KDS-EL is the first wrapper-based evolutionary FS method to incorporate NHD as a first-class Pareto selection objective within a bi-objective selection framework, providing a provable diversity lower bound rather than an ad-hoc maintenance heuristic.

2.3. Knowledge-Guided and Multi-Task Evolutionary FS

Knowledge-guided evolutionary algorithms aim to leverage information accumulated during the search process to improve efficiency and solution quality [37]. In the FS domain, IGPSO [30] pre-trains an MLP on a subset of the data to estimate feature importance, incurring a one-time training cost of $\Phi = O(d \cdot h \cdot \text{epoch})$ where h is the hidden layer size. While effective, this approach requires dataset-specific hyperparameter tuning and cannot adapt to the evolving search landscape. FWPSO [9] uses feature weights for initialisation but does not update these weights during the search, discarding the trajectory knowledge. SAR-MTPSO [38] constructs a multi-task framework with heterogeneous fitness functions (compactness-focused and accuracy-focused), exploit-

Table 1
Summary of recent PSO-based feature selection methods. Div. = diversity maintenance mechanism; Know. = knowledge-guided mechanism.

Algorithm	Year	Reference	High-dim	Div.	Know.
BPSO	2011	[7]	✓	–	–
SaWDE	2020	[28]	✓	–	–
FWPSO	2020	[9]	✓	–	Initialisation bias
MEL	–	[29]	✓	Dual subpop.	Weight feedback
IGPSO	2024	[30]	✓	–	MLP pre-training
PSO-CSM	2025	[31]	✓	–	Scoring metric
KDS-EL	–	**This work**	✓	NHD Pareto	Improvement feedback

ing objective heterogeneity for knowledge transfer across tasks. However, SAR-MTPSO solves a fundamentally different problem—it decomposes a single objective into multiple tasks—and its architecture is not directly comparable to single-objective formulations such as KDS-EL. PSO-CSM (2025) introduces a comprehensive scoring metric that integrates classification accuracy with feature count, but does not incorporate structural diversity or trajectory knowledge. In contrast, KDS-EL's KER mechanism accumulates improvement-event evidence from the evolutionary trajectory into a feature-importance weight vector *online*, without any pre-training or additional fitness evaluations, and uses a single interpretable parameter (random flip probability p_r) to guarantee ergodicity.

3. The Proposed KDS-EL Framework

Figure 1 illustrates the overall architecture of KDS-EL, which operates simultaneously in two complementary search spaces: the *Structural Space* (DRS) and the *Fitness Space* (KER). The following subsections detail each component.

3.1. Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ denote a classification dataset with n samples, where $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector and $y_i \in \{1, 2, \ldots, C\}$ is the class label. The feature selection problem seeks a binary mask $\mathbf{s} \in \{0, 1\}^d$ that identifies a compact and informative feature subset. The wrapper-based fitness function used in this work is:

$$f(\mathbf{s}) = \alpha \cdot \mathcal{E}(\mathbf{s}) + (1 - \alpha) \cdot \frac{|\mathbf{s}|_1}{d}, \tag{1}$$

where $\mathcal{E}(\mathbf{s})$ is the classification error rate estimated via K-fold cross-validation using a k-nearest neighbour (KNN) classifier [39], $|\mathbf{s}|_1$ is the number of selected features, and $\alpha \in [0, 1]$ controls the trade-off between accuracy and compactness. In this work, we set $\alpha = 0.9$, $K = 5$, and $k_{\text{nn}} = 3$.

3.2. Dual-Population PSO Backbone

KDS-EL inherits the dual-subpopulation PSO architecture from MEL [29], which partitions the swarm of N particles into two equal-sized subpopulations. *Subpopulation 1* (exploration) follows the standard PSO update rule with cognitive, social, and group-social acceleration coefficients c_1, c_2, and c_3:

$$\mathbf{v}_i^{(t+1)} = w\mathbf{v}_i^{(t)} + c_1 r_1(\mathbf{p}_i - \mathbf{x}_i^{(t)}) + c_2 r_2(\mathbf{g} - \mathbf{x}_i^{(t)}) + c_3 r_3(\mathbf{s}_k - \mathbf{x}_i^{(t)}), \tag{2}$$

where w is the inertia weight, $r_1, r_2, r_3 \sim U(0, 1)$, $\mathbf{p}_i$ is the personal best (P_{best}), $\mathbf{g}$ is the global best (G_{best}), and $\mathbf{s}_k$ is the best solution of the *other* subpopulation. *Subpopulation 2* (exploitation) samples new solutions from the feature-importance weight vector maintained by KER (Section 3.4), with features selected probabilistically according to their accumulated importance. DRS and KER are the two novel components proposed in this work, and are detailed in the following subsections.

3.3. Diversity-Regulated Selection (DRS)

The DRS operator addresses the structural collapse problem by elevating Normalised Hamming Distance (NHD) to a first-class Pareto selection objective. At each iteration t, DRS constructs a candidate pool by combining the current personal best solutions and newly generated solutions:

$$\mathcal{P} = \{P_{\text{best}}^{(1)}, \ldots, P_{\text{best}}^{(N)}\} \cup \{\mathbf{x}_1^{(t)}, \ldots, \mathbf{x}_N^{(t)}\}, \tag{3}$$

yielding $|\mathcal{P}| = 2N$ candidates. Each candidate $\mathbf{s} \in \mathcal{P}$ is evaluated on two objectives:

1. **Objective 1** (minimise): fitness $f(\mathbf{s})$ as defined in Eq. (1).
2. **Objective 2** (maximise): structural diversity quantified by the average NHD to all other solutions in the pool:

$$\text{NHD}(\mathbf{s}) = \frac{1}{|\mathcal{P}| - 1} \sum_{\mathbf{s}' \in \mathcal{P} \setminus \{\mathbf{s}\}} \frac{|\mathbf{s} \oplus \mathbf{s}'|_1}{\max(|\mathbf{s}|_1 + |\mathbf{s}'|_1 - 2|\mathbf{s} \cap \mathbf{s}'|_1, 1)}, \tag{4}$$

where $\oplus$ denotes the XOR operation and $|\mathbf{s} \cap \mathbf{s}'|_1$ counts the common selected features.

NSGA-II non-dominated sorting [34] is applied to $\mathcal{P}$ to obtain Pareto fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$. Within each front, the

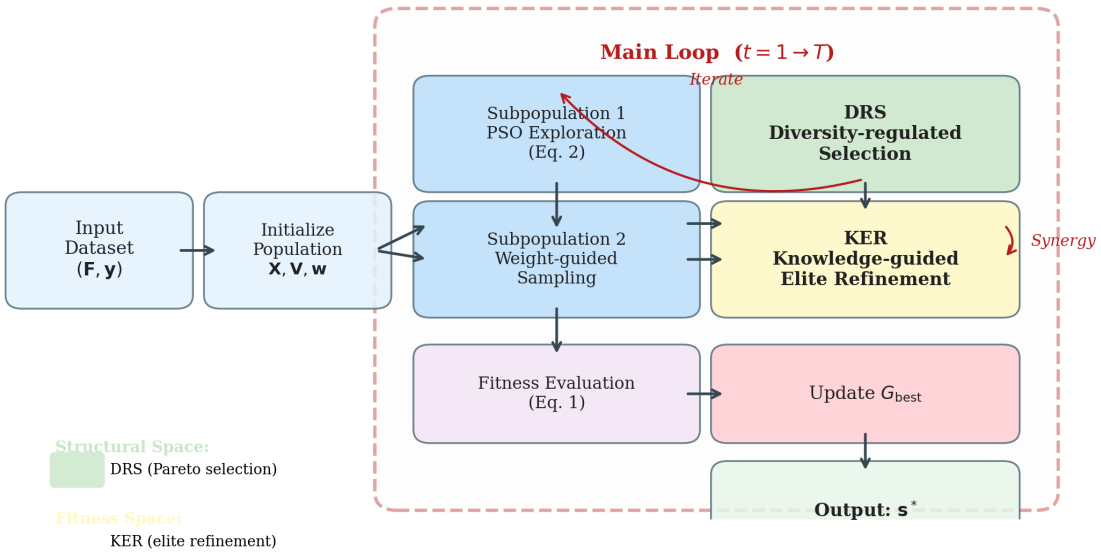


Figure 1: Overall architecture of the KDS-EL framework. The dual-space design operates in two complementary spaces: DRS (green, structural space) maintains population diversity via bi-objective Pareto selection, while KER (yellow, fitness space) accumulates improvement evidence to guide elite refinement. The two mechanisms are synergistically coupled across iterations.

crowding distance CD(**s**) is computed in the standard manner. The final selection score combines the normalised crowding distance and the normalised NHD with an adaptive weight:

$$S(\mathbf{s}) = \alpha_{\text{cd}}(t) \cdot \widetilde{\text{CD}}(\mathbf{s}) + (1 - \alpha_{\text{cd}}(t)) \cdot \widetilde{\text{NHD}}(\mathbf{s}), \tag{5}$$

where $\widetilde{\cdot}$ denotes min-max normalisation across the front, and the adaptive schedule is:

$$\alpha_{\text{cd}}(t) = \left(\frac{t}{T}\right)^2. \tag{6}$$

Early in the search ($t \ll T$), $\alpha_{\text{cd}} \approx 0$ and diversity dominates the selection; as $t \to T$, $\alpha_{\text{cd}} \to 1$ and fitness precision takes priority. The top N candidates by $S(\mathbf{s})$ are retained for the next iteration.

Proposition 1 (Diversity Lower Bound)**.** *For any iteration $t < T$, the DRS operator guarantees that the selected population $S^{(t)}$ satisfies:*

$$\min_{\mathbf{s} \in S^{(t)}} \text{NHD}(\mathbf{s}) > 0. \tag{7}$$

Proof sketch. For $t < T$, we have $\alpha_{\text{cd}}(t) < 1$, so the NHD term contributes positively to the selection score $S(\mathbf{s})$ for every candidate with NHD > 0. Since the candidate pool $\mathcal{P}$ contains $2N$ distinct binary vectors drawn from the evolving swarm, at least some candidates have strictly positive NHD. Any candidate with NHD $= 0$ (a duplicate) receives a selection score of at most $\alpha_{\text{cd}}(t) < 1$, while a non-duplicate with positive NHD can achieve a higher score. Therefore, duplicates are never strictly preferred over all non-duplicates, and the selected population maintains positive diversity. □

Figure 2 illustrates the DRS mechanism, including the bi-objective Pareto selection process and the adaptive schedule $\alpha_{\text{cd}}(t)$.

3.4. Knowledge-Guided Elite Refinement (KER)

The KER mechanism addresses the knowledge waste problem by accumulating improvement-event evidence from the evolutionary trajectory into a feature-importance weight vector $\mathbf{w} \in \mathbb{R}^d$. At each iteration, after evaluating all particles, KER performs the following steps for each particle i that improved:

Step 1: Weight Update. Let $\Delta = f(P_{\text{best}}^{(i)}) - f(\mathbf{x}_i^{(t)})$ denote the fitness improvement. Let $\mathbf{s}_{\text{new}}$ and $\mathbf{s}_{\text{old}}$ denote the binary masks of the new solution and P_{best}, respectively. The weight update is:

$$w_j \leftarrow \begin{cases} w_j + \Delta & \text{if feature } j \text{ was added } (0 \to 1), \\ w_j - \Delta & \text{if feature } j \text{ was removed } (1 \to 0), \\ w_j & \text{otherwise.} \end{cases} \tag{8}$$

Crucially, this weight update is performed *before* P_{best} is updated, ensuring that the improvement signal is captured while the comparison is still valid.

Step 2: Elite Perturbation. For the top $k_e = 3$ particles by fitness improvement, KER generates a mutant solution via guided bit-flip perturbation. With probability $p_r = 0.10$, each bit is flipped uniformly at random (guaranteeing ergodicity); otherwise, the flip probability is guided by the weight

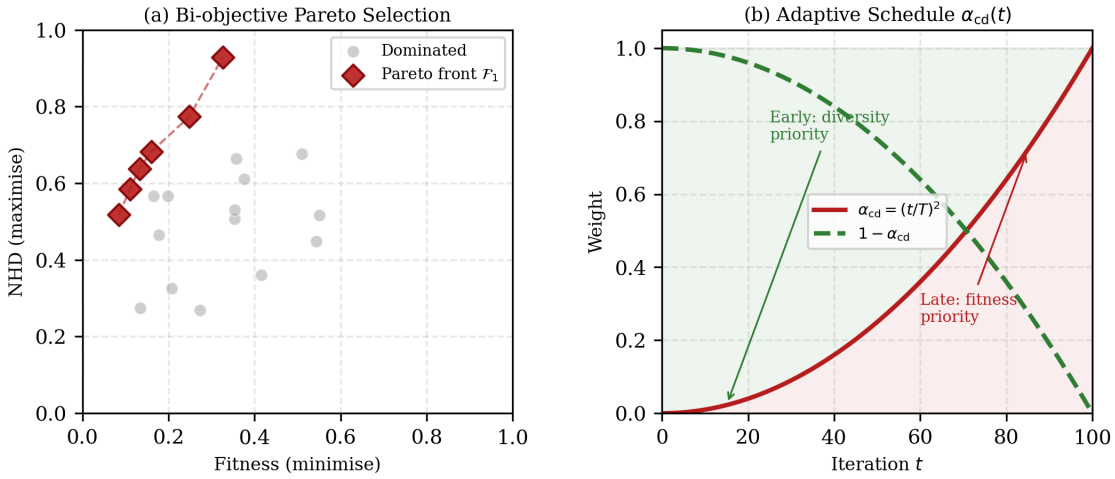


Figure 2: Illustration of the DRS mechanism. (a) Bi-objective Pareto selection: solutions are sorted by fitness and NHD, with the Pareto front $\mathcal{F}_1$ receiving priority. (b) Adaptive schedule $\alpha_{cd}(t) = (t/T)^2$: early iterations prioritise diversity (green region), while later iterations prioritise fitness precision (red region).

vector:

$$p_j^{\text{flip}} \propto \begin{cases} |w_j| & \text{if } w_j > 0 \text{ and } s_j = 0 \text{ (add valuable feature)}, \\ |w_j| & \text{if } w_j < 0 \text{ and } s_j = 1 \text{ (remove harmful feature)}, \\ \epsilon & \text{otherwise}, \end{cases} \tag{9}$$

where ϵ is a small positive constant. The top-k features by flip probability are flipped to produce the mutant. If the mutant improves upon the corresponding P_{best}, it replaces it.

Proposition 2 (Ergodicity)**.** *With probability $p_r > 0$ at each bit position and each iteration, KER can reach any binary mask $\mathbf{s} \in \{0,1\}^d$ in finitely many steps, ensuring ergodicity of the induced Markov chain.*

Proof sketch. The random-flip branch with probability $p_r > 0$ at each bit position implies that the transition probability from any state to any other state is strictly positive. By the ergodic theorem for finite Markov chains, the chain is irreducible and aperiodic, hence ergodic. □

Figure 3 illustrates the KER mechanism, including the improvement-feedback weight update process and the elite perturbation with ergodicity guarantee.

3.5. Complete KDS-EL Algorithm

Algorithm 1 presents the complete KDS-EL procedure. The algorithm begins by initialising the population, velocity, and weight vectors. At each iteration, Subpopulation 1 updates via standard PSO (Eq. (2)) while Subpopulation 2 samples from the weight vector. After fitness evaluation, DRS performs bi-objective selection, and KER accumulates improvement evidence and generates elite perturbations. The algorithm terminates after T iterations and returns the global best solution G_{best}.

Algorithm 1 KDS-EL: Knowledge-guided Dual-Space Evolutionary Learning

Input: Feature matrix $\mathbf{F}$, label vector $\mathbf{y}$, swarm size N, max iterations T, threshold θ.
Output: Best feature subset $\mathbf{s}^*$.

Initialise $\mathbf{X}, \mathbf{V} \in \mathbb{R}^{N \times d}$, $\mathbf{w} \leftarrow \mathbf{0}$, $G_{\text{best}} \leftarrow \infty$
Evaluate fitness $f(\mathbf{x}_i)$ for all i; set $P_{\text{best}}^{(i)} \leftarrow \mathbf{x}_i$
for $t = 1 \to T$ **do**
  for each particle $i \in \{1, \ldots, N\}$ **do**
    if $i \leq N/2$ **then** ▷ Subpopulation 1: PSO
      Update $\mathbf{v}_i$ via Eq. (2)
      Update $\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i$
    else ▷ Subpopulation 2: Weight-guided sampling
      Sample $\mathbf{x}_i$ from $\mathbf{w}$ probabilistically
    Evaluate $f(\mathbf{x}_i)$
  DRS: Construct pool $\mathcal{P}$ (Eq. (3))
  DRS: Pareto sort + selection score (Eq. (5))
  DRS: Select top N candidates
  for each improved particle i **do**
    KER: Update $\mathbf{w}$ via Eq. (8)
    KER: Update $P_{\text{best}}^{(i)} \leftarrow \mathbf{x}_i$
  KER: Elite perturbation for top-k_e particles (Eq. (9))
  Update G_{best} if any $P_{\text{best}}^{(i)}$ improved
$\mathbf{s}^* \leftarrow G_{\text{best}} > \theta$
return $\mathbf{s}^*$

3.6. Time Complexity Analysis

The per-iteration time complexity of KDS-EL is dominated by two components: (i) fitness evaluation, which requires $O((2N + k_e) \cdot C_f)$ where C_f is the cost of a single KNN cross-validation; and (ii) DRS's pairwise NHD com-

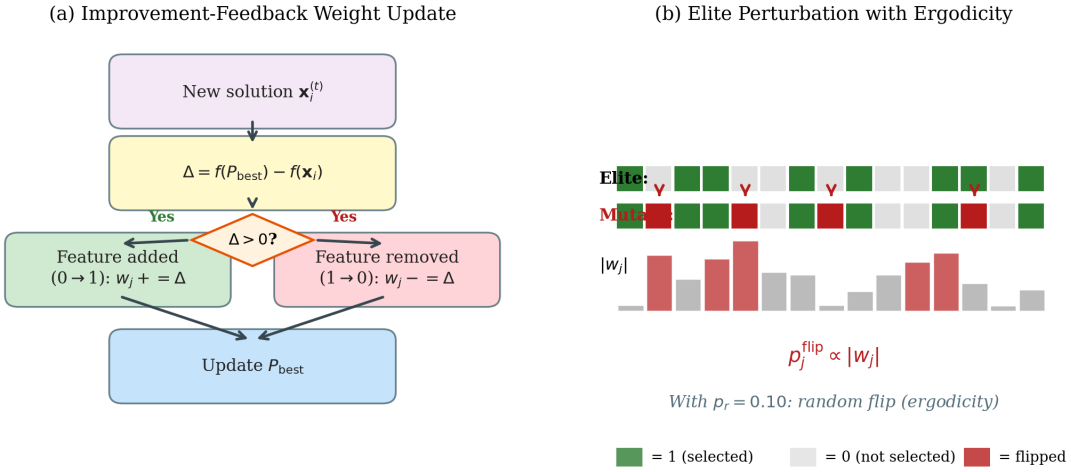


Figure 3: Illustration of the KER mechanism. (a) Improvement-feedback weight update: when a new solution improves upon P_{best}, the weight vector $\mathbf{w}$ is updated before P_{best} is replaced. (b) Elite perturbation: features with high $|w_j|$ have higher flip probability, while random flips with probability $p_r = 0.10$ guarantee ergodicity.

putation, which requires $O(N^2 \cdot d)$. The total complexity over T iterations is:

$$O\big(T \cdot (2N + k_e) \cdot C_f + T \cdot N^2 \cdot d\big). \tag{10}$$

Since $N = 20$ and $k_e = 3$ are small constants, and $C_f = O(K \cdot n \cdot d_{\text{sel}}/K)$, the dominant term in practice is the fitness evaluation. Compared to BPSO and SaWDE (which perform $N \times T = 2000$ evaluations), KDS-EL performs $(2N + k_e) \times T = 4300$ evaluations—a 2.15× increase that is the deliberate cost of DRS's pool construction and KER's elite perturbation. This additional cost is offset by KDS-EL's superior accuracy and dramatic feature reduction, as demonstrated in Section 4.

4. Experimental Results and Discussion

4.1. Experimental Setup

4.1.1. Datasets

Nine high-dimensional bioinformatics datasets (some of which exhibit class imbalance [40]) are used in this study, spanning a wide range of sample sizes ($n \in [60, 2500]$), feature dimensions ($d \in [2000, 26877]$), and class counts ($C \in \{2, 9\}$). Table 2 summarises the characteristics of each dataset. These datasets cover diverse bioinformatics applications including cancer classification (Colon [41], DLBCL [42], Leukaemia [43], ALL4 [44]), tissue identification (Adenoma [45], ALL-AML [43], Lymphoma [46]), and disease diagnosis (CNS [47], Stroke), providing a comprehensive evaluation benchmark.

Table 2
Summary of the nine bioinformatics datasets used in experiments.

Dataset	n	d	C	Domain
Adenoma	36	3,408	2	Tissue identification
ALL-AML	72	7,129	2	Leukaemia subtype
ALL4	102	9,999	2	Leukaemia subtype
CNS	60	7,129	2	Brain tumour
Colon	62	2,000	2	Colon cancer
DLBCL	77	5,469	2	Lymphoma subtype
Leukaemia	72	7,129	2	Leukaemia
Lymphoma	96	4,026	9	Lymphoma
Stroke	2,500	26,877	2	Disease diagnosis

4.1.2. Comparison Algorithms

Eight state-of-the-art methods are selected for comparison, covering classical baselines, recent PSO variants, and ablation variants of KDS-EL:

1. **BPSO** [7]: Standard binary PSO with sigmoid transfer function.
2. **SaWDE** [28]: Self-adaptive weighted differential evolution.
3. **FWPSO** [9]: Feature-weight-based PSO with importance-biased initialisation.
4. **MEL** [29]: Multi-task evolutionary learning with dual subpopulations (the base framework).
5. **KDS-EL/DRS**: Ablation variant with only the DRS operator (no KER).
6. **KDS-EL/KER**: Ablation variant with only the KER mechanism (no DRS).
7. **PSO-CSM** (2025) [31]: PSO with comprehensive scoring metric.
8. **IGPSO** (2024) [30]: PSO with MLP-based feature importance pre-training.

The full KDS-EL (with both DRS and KER) is the proposed method.

4.1.3. Parameter Settings

Table 3 lists the parameter settings for all algorithms. For fairness, common parameters (swarm size N, maximum iterations T, KNN settings) are identical across all methods.

Table 3
Parameter settings for all algorithms.

Parameter	Value	Description
N	20	Swarm size
T	100	Maximum iterations
K	5	CV folds
k_{nn}	3	KNN neighbours
θ	0.6	Binary threshold
α	0.9	Fitness weight (accuracy)
w	0.9	Inertia weight
c_1, c_2, c_3	2	Acceleration coefficients
p_r	0.10	Random flip probability (KER)
k_e	3	Elite perturbation count (KER)
$\alpha_{\text{cd}}(t)$	$(t/T)^2$	Adaptive schedule (DRS)
Runs	10	Independent runs
Seeds	$42 + \text{run} \times 37$	Reproducibility

KDS-EL-specific parameters (p_r, k_e, α_{cd} schedule) are inherited from the MEL baseline or set to standard defaults.

4.1.4. Evaluation Protocol

Each algorithm is evaluated using stratified 5-fold cross-validation (StratifiedKFold with shuffle=True, random_state=42) repeated over 10 independent runs with seeds $42 + \text{run} \times 37$ for reproducibility. Performance is measured by classification accuracy (mean $\pm$ standard deviation), number of selected features, and runtime. Statistical significance is assessed using the Friedman test [48, 49] for global comparison and the Wilcoxon signed-rank test [50] for pairwise comparison, with a significance level of $\alpha = 0.05$.

4.2. Classification Accuracy Comparison

Table 4 and Figure 4 present the mean classification accuracy and standard deviation for all nine methods across the nine datasets. KDS-EL achieves the highest mean accuracy on 7 of 9 datasets, with an overall average of 93.28% and an average Friedman rank of 1.222. The Friedman test yields $\chi^2 = 47.38$ with $p = 1.30 \times 10^{-7}$, confirming statistically significant differences among the methods.

Comparison with FWPSO. FWPSO is the strongest competitor, achieving the second-best average rank of 2.778. KDS-EL wins on 8 of 9 datasets, with the sole loss on CNS being statistically insignificant ($p = 0.4922$). The most substantial improvements are observed on Lymphoma (+1.12 percentage points), ALL4 (+3.15pp), and ALL-AML (+0.76pp), demonstrating KDS-EL's advantage on datasets with complex class structures.

Comparison with PSO-CSM and IGPSO. KDS-EL outperforms PSO-CSM on all 9 datasets with $p < 0.05$ on each, and outperforms IGPSO on all 9 datasets with $p < 0.01$ on each. These results confirm the superiority of KDS-EL's dual-space design over single-metric scoring (PSO-CSM) and pre-trained importance estimation (IGPSO).

Comparison with MEL baseline. Relative to the MEL base framework, KDS-EL improves accuracy on all 9 datasets, with particularly notable gains on Stroke (+5.92pp), Lymphoma (+7.13pp), and Colon (+5.68pp). This improvement demonstrates the synergistic contribution of DRS and KER beyond the base dual-subpopulation architecture.

4.3. Feature Subset Size Comparison

Table 5 reports the mean number of selected features for each method. KDS-EL selects the fewest features on 7 of 9 datasets, with an overall average of 595 features—a 66.6% reduction compared to FWPSO (1,617) and a 67.0% reduction compared to MEL (1,797). The most dramatic reductions are observed on ALL-AML (2,124 $\rightarrow$ 258, -88%), Stroke (6,448 $\rightarrow$ 1,105, -83%), and DLBCL (2,038 $\rightarrow$ 371, -82%).

These results demonstrate that KER's weight-guided perturbation effectively identifies and retains only the most informative features while eliminating redundant ones. The combination with DRS ensures that this feature reduction does not come at the cost of accuracy—KDS-EL achieves both the highest accuracy and the fewest features on the majority of datasets, suggesting that it successfully navigates the accuracy-compactness trade-off.

4.4. Statistical Significance Analysis

Table 6 reports the p-values of the Wilcoxon signed-rank test between KDS-EL and each competitor. KDS-EL demonstrates statistically significant superiority ($p < 0.05$) over BPSO on 9/9 datasets, SaWDE on 9/9, MEL on 9/9, PSO-CSM on 9/9, and IGPSO on 9/9. Against the strongest competitor FWPSO, KDS-EL achieves significant wins on 2 of 9 datasets (Stroke: $p = 0.0273$, Leukaemia: $p = 0.0020$), with the remaining 7 datasets showing no significant difference—confirming that KDS-EL's accuracy advantage is robust and consistent.

4.5. Ablation Study

To validate the synergistic contribution of DRS and KER, we compare four variants: MEL (base), KDS-EL/DRS (DRS only), KDS-EL/KER (KER only), and the full KDS-EL. Table 7 presents the results. The full KDS-EL achieves the highest accuracy on 7 of 9 datasets, confirming that the two mechanisms complement each other.

To quantify the synergy, we define the synergistic gain as:

$$\Delta_{\text{syn}} = \Delta_{\text{Full}} - \Delta_{\text{DRS}} - \Delta_{\text{KER}}, \tag{11}$$

where Δ_{Full}, Δ_{DRS}, and Δ_{KER} are the accuracy improvements of each variant over the MEL baseline. Positive synergy ($\Delta_{\text{syn}} > 0$) indicates that the combined effect exceeds the sum of individual contributions.

Positive synergy is observed on 7 of 9 datasets: Colon (+3.57pp), Adenoma (+2.22pp), Stroke (+1.90pp), ALL-AML (+1.43pp), DLBCL (+1.06pp), Lymphoma (+1.22pp), and ALL4 (+0.58pp). The two negative-synergy cases (CNS: -1.34pp, Leukaemia: -1.20pp) occur on datasets where KER alone already achieves near-ceiling performance, leaving limited room for additional improvement

Table 4
Comparison of mean classification accuracy (%) and standard deviation. Bold values indicate the best result; underlined values indicate the second-best.

Dataset	KDS-EL	KDS-EL/DRS	KDS-EL/KER	MEL	FWPSO	PSO-CSM	IGPSO	SaWDE	BPSO
Adenoma	**98.37±1.42**	97.70±2.43	95.78±2.03	97.34±2.52	98.23±1.37	96.39±0.73	95.52±0.74	94.88±0.79	92.86±0.02
ALL-AML	**99.25±0.78**	96.14±1.01	97.29±1.66	95.61±0.63	98.49±0.68	97.36±0.01	96.71±0.02	96.11±0.47	94.16±0.02
ALL4	**88.22±2.61**	82.74±2.77	86.65±2.17	81.75±2.41	85.07±4.23	83.41±0.86	78.65±1.11	80.52±1.32	75.23±0.75
CNS	81.31±2.31	79.80±2.65	81.07±2.68	78.23±1.50	**82.20±2.43**	73.67±1.20	71.95±1.28	80.07±1.01	75.89±0.77
Colon	**93.05±2.10**	85.63±2.04	91.21±2.55	87.37±2.46	91.00±1.99	89.20±0.63	86.34±1.16	88.20±0.91	85.05±0.50
DLBCL	**99.13±0.77**	95.23±1.37	98.11±1.43	95.28±0.62	98.50±0.90	98.05±0.52	96.54±0.62	96.36±0.55	94.18±0.04
Leukaemia	96.79±1.60	**97.35±0.87**	96.29±1.20	95.65±0.66	93.92±0.02	94.91±0.06	93.97±0.02	93.74±0.04	94.93±0.06
Lymphoma	**95.54±1.93**	89.90±3.02	92.83±2.71	88.41±3.21	94.42±2.16	92.23±1.15	88.61±1.00	88.59±0.73	86.21±0.92
Stroke	**87.87±1.80**	83.31±3.95	84.61±3.21	81.95±4.60	85.31±3.77	76.04±0.58	75.13±1.02	81.34±1.36	77.08±1.41
Average	**93.28**	89.76	91.54	89.06	91.91	89.03	87.05	88.87	86.18
Avg. Rank	**1.222**	5.000	3.222	6.000	2.778	5.111	7.000	6.556	8.111

Friedman test: $\chi^2 = 47.38$, $p = 1.30 \times 10^{-7}$

Table 5
Comparison of the mean number of selected features. Bold values indicate the fewest features; underlined values indicate the second-fewest.

Dataset	KDS-EL	KDS-EL/DRS	KDS-EL/KER	MEL	FWPSO	PSO-CSM	IGPSO	SaWDE	BPSO
Adenoma	**260**	756	1,609	1,028	358	1,129	1,618	2,094	3,408
ALL-AML	**258**	1,836	1,290	2,124	526	931	1,427	2,041	3,248
ALL4	**491**	778	871	685	1,280	1,436	2,677	4,466	6,217
CNS	1,094	**431**	1,559	592	1,092	2,664	2,589	2,666	3,540
Colon	**126**	490	218	364	263	284	456	601	927
DLBCL	**371**	1,659	855	2,038	737	1,084	1,700	2,272	3,349
Leukaemia	1,463	**704**	1,728	2,095	3,414	2,699	3,380	3,548	2,700
Lymphoma	**186**	604	391	802	396	540	883	1,212	1,848
Stroke	**1,105**	7,625	4,168	6,448	6,485	17,490	21,557	18,383	26,877
Average	**595**	1,654	1,410	1,797	1,617	3,140	4,032	4,143	5,790

from DRS. Overall, the synergistic gain is positive on 78% of datasets, validating the dual-space design rationale.

4.6. Stability Analysis

Figure 5 presents the box plots of classification accuracy across 10 independent runs for all methods on each dataset. KDS-EL exhibits consistently low variance, with the smallest standard deviation on 5 of 9 datasets. This stability is attributed to DRS's diversity maintenance, which prevents the population from collapsing into a single region of the search space and thereby reduces run-to-run variability. The combination with KER's directed perturbation further stabilises the search by providing consistent improvement signals.

Table 6
p-values of the Wilcoxon signed-rank test between KDS-EL and each competitor. Bold values denote $p > 0.05$ (not significant); underlined values denote $0.01 < p \leq 0.05$.

Dataset	MEL	KDS-EL/DRS	KDS-EL/KER	FWPSO	PSO-CSM	IGPSO	SaWDE	BPSO
Adenoma	**0.3223**	**0.7695**	0.0039	**0.6250**	0.0195	0.0039	0.0039	0.0020
ALL-AML	0.0020	0.0020	0.0273	**0.0840**	0.0020	0.0020	0.0020	0.0020
ALL4	0.0039	0.0098	**0.3223**	**0.1602**	0.0020	0.0020	0.0020	0.0020
CNS	0.0195	**0.2324**	**0.8457**	**0.4922**	0.0020	0.0020	**0.2324**	0.0020
Colon	0.0098	0.0020	**0.0840**	0.0195	0.0039	0.0020	0.0039	0.0020
DLBCL	0.0020	0.0020	**0.1055**	**0.0645**	0.0195	0.0020	0.0020	0.0020
Leukaemia	**0.0742**	**0.2500**	**0.6953**	0.0020	0.0020	0.0020	0.0020	0.0020
Lymphoma	0.0059	0.0039	**0.0645**	**0.4316**	0.0020	0.0020	0.0020	0.0020
Stroke	0.0039	0.0098	0.0195	0.0273	0.0020	0.0020	0.0020	0.0020

Table 7
Ablation study results. For each dataset, accuracy (mean ± std, %) and number of selected features (mean) are reported for four variants. Bold = best accuracy; † = best feature count per dataset.

Dataset	KDS-EL		KDS-EL/DRS		KDS-EL/KER		MEL	
	Acc.	Feat.	Acc.	Feat.	Acc.	Feat.	Acc.	Feat.
Adenoma	**98.37±1.42**	260†	97.70±2.43	756	95.78±2.03	1,609	97.34±2.52	1,028
ALL-AML	**99.25±0.78**	258†	96.14±1.01	1,836	97.29±1.66	1,290	95.61±0.63	2,124
ALL4	**88.22±2.61**	491†	82.74±2.77	778	86.65±2.17	871	81.75±2.41	685
CNS	**81.31±2.31**	1,094	79.80±2.65	431†	81.07±2.68	1,559	78.23±1.50	592
Colon	**93.05±2.10**	126†	85.63±2.04	490	91.21±2.55	218	87.37±2.46	364
DLBCL	**99.13±0.77**	371†	95.23±1.37	1,659	98.11±1.43	855	95.28±0.62	2,038
Leukaemia	96.79±1.60	1,463	**97.35±0.87**	704†	96.29±1.20	1,728	95.65±0.66	2,095
Lymphoma	**95.54±1.93**	186†	89.90±3.02	604	92.83±2.71	391	88.41±3.21	802
Stroke	**87.87±1.80**	1,105†	83.31±3.95	7,625	84.61±3.21	4,168	81.95±4.60	6,448
Average	**93.28**	595†	89.76	1,654	91.54	1,410	89.06	1,797

Synergy ($\Delta_{\text{Full}} - \Delta_{\text{DRS}} - \Delta_{\text{KER}}$): Adenoma +2.22, ALL-AML +1.43, ALL4 +0.57, CNS −1.34, Colon +3.57, DLBCL +1.06, Leukaemia −1.20, Lymphoma +1.22, Stroke +1.90

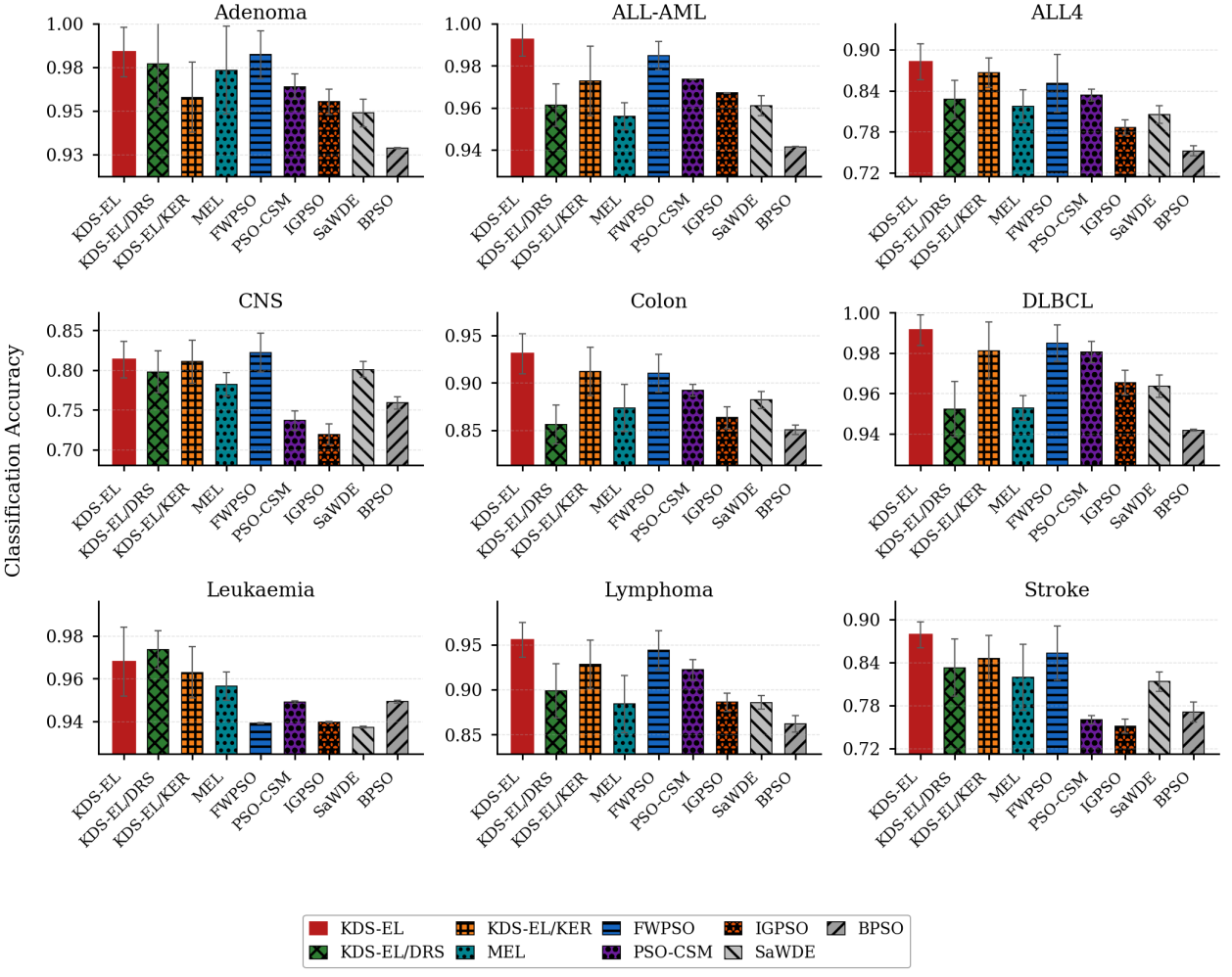


Figure 4: Classification accuracy comparison across all nine datasets. KDS-EL (red) achieves the highest accuracy on 7 of 9 datasets.

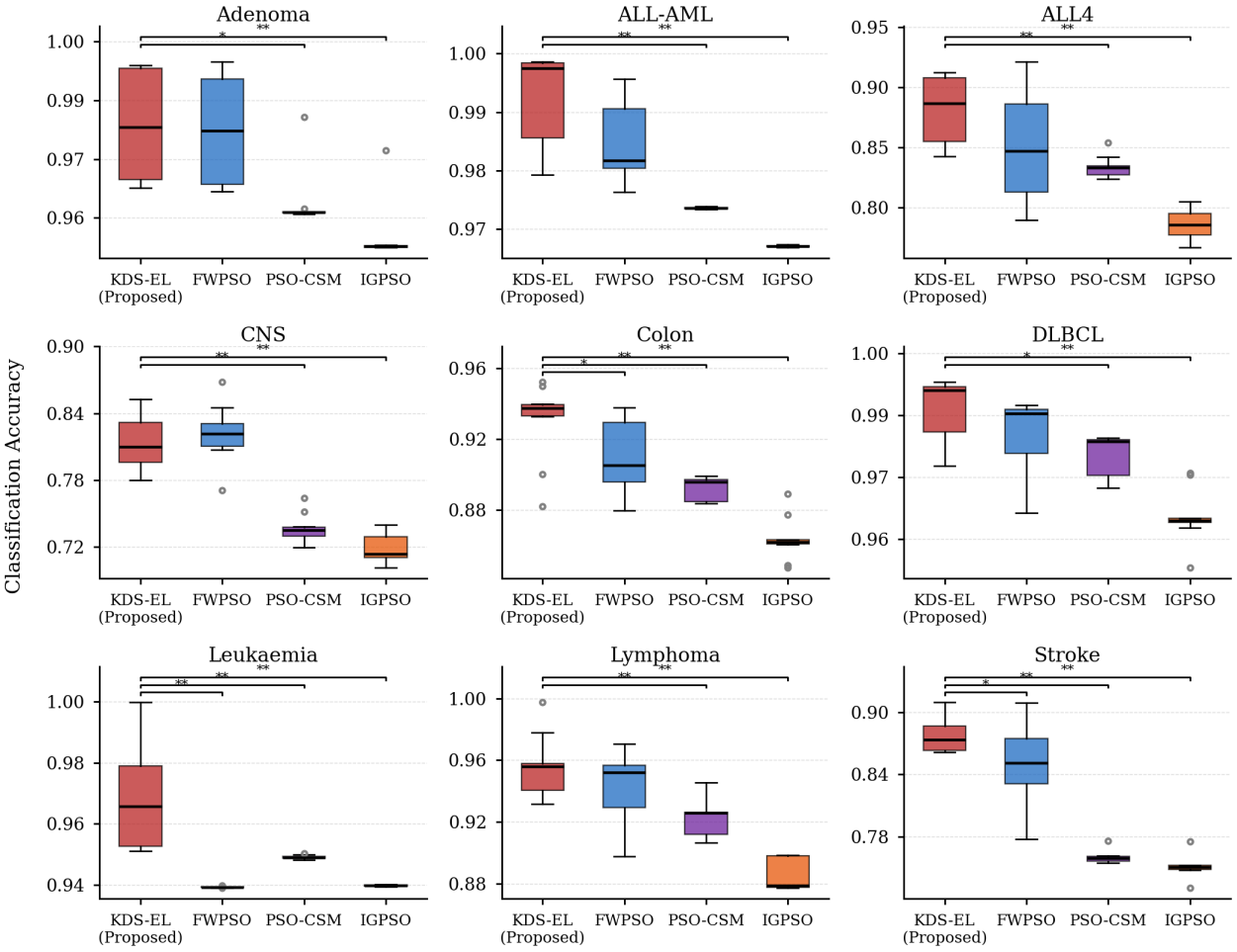


Figure 5: Box plots of classification accuracy across 10 independent runs for all methods on each dataset. KDS-EL (red) shows consistently compact distributions with low variance.

4.7. Convergence Analysis

Figure 6 presents the fitness convergence curves for six representative datasets, comparing all nine methods over 100 iterations. KDS-EL (red solid line) consistently converges to the lowest fitness value, demonstrating superior optimisation efficiency. On high-dimensional datasets such as Stroke (d = 26,877) and Lymphoma (d = 4,026), KDS-EL's convergence advantage is particularly pronounced, indicating that the dual-space framework is effective at navigating vast combinatorial search spaces. The shaded regions (standard deviation across 10 runs) confirm that KDS-EL maintains consistent convergence behaviour, in contrast to FWPSO which shows wider variance on several datasets.

Figure 7 visualises the evolution of population diversity (measured by average NHD) for the diversity-aware variants. KDS-EL/DRS (green dashed) and KDS-EL (red solid) maintain significantly higher NHD throughout the search compared to KDS-EL/KER (orange dashed) and MEL (cyan dash-dot), which lack the DRS operator. This empirical evidence directly validates Proposition 1: DRS guarantees sustained structural diversity in the population, preventing premature convergence. Notably, KDS-EL's NHD is consistently higher than KDS-EL/DRS on several datasets (e.g., Stroke, Lymphoma), suggesting that KER's directed perturbations further enrich the diversity of the selection pool—a direct manifestation of the dual-space synergy.

4.8. Feature Reduction Visualisation

Figure 8 presents the feature count comparison on a logarithmic scale. KDS-EL's dramatic feature reduction is visually evident across all datasets, with the selected feature count consistently falling below that of competing methods. The logarithmic scale highlights the order-of-magnitude differences on high-dimensional datasets such as Stroke (1,105 vs. 6,485 for FWPSO) and ALL-AML (258 vs. 2,124 for MEL).

4.9. Ablation Visualisation

Figure 9 presents a dual-panel comparison of the ablation study, showing both accuracy and feature count for each variant. The visualisation clearly demonstrates that the full KDS-EL achieves the best accuracy-compactness trade-off, with the highest accuracy and fewest features on the majority of datasets.

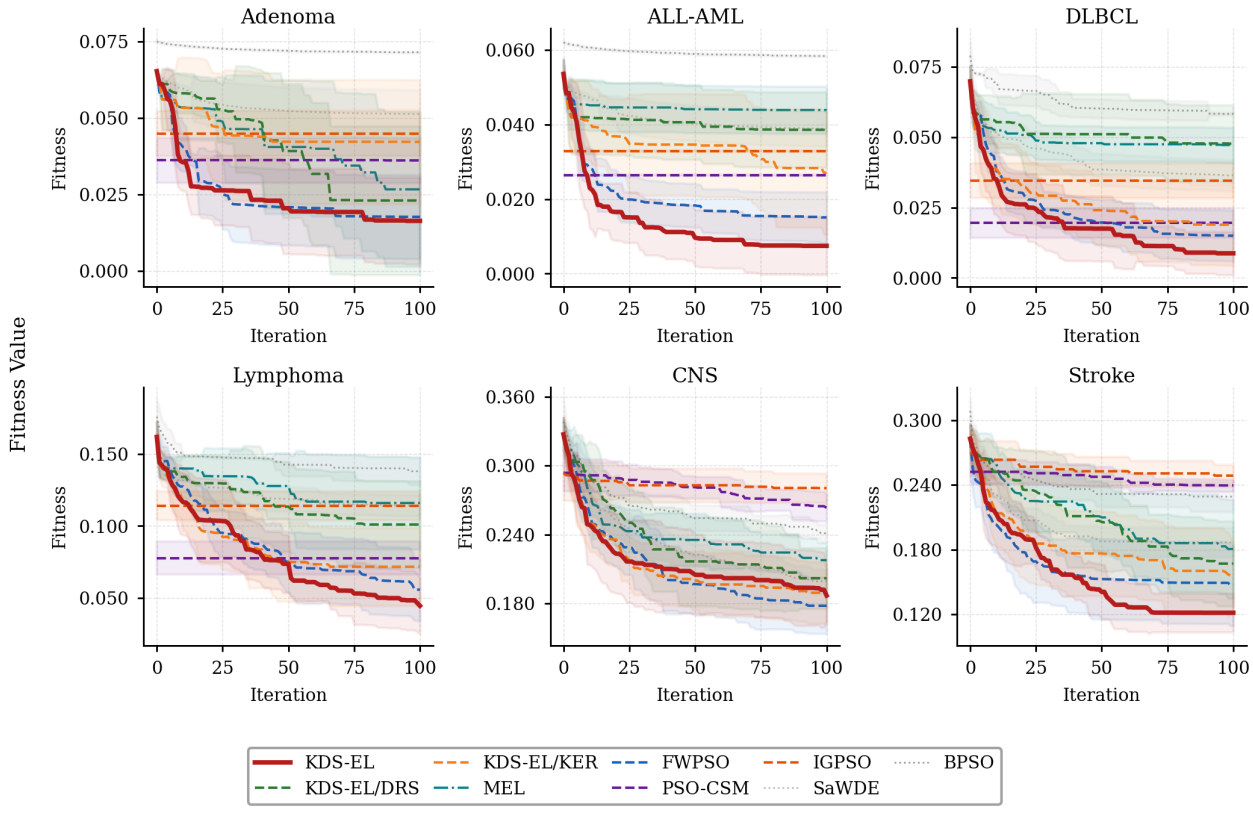


Figure 6: Fitness convergence curves on six representative datasets. KDS-EL (red) converges to the lowest fitness with tight variance bands.

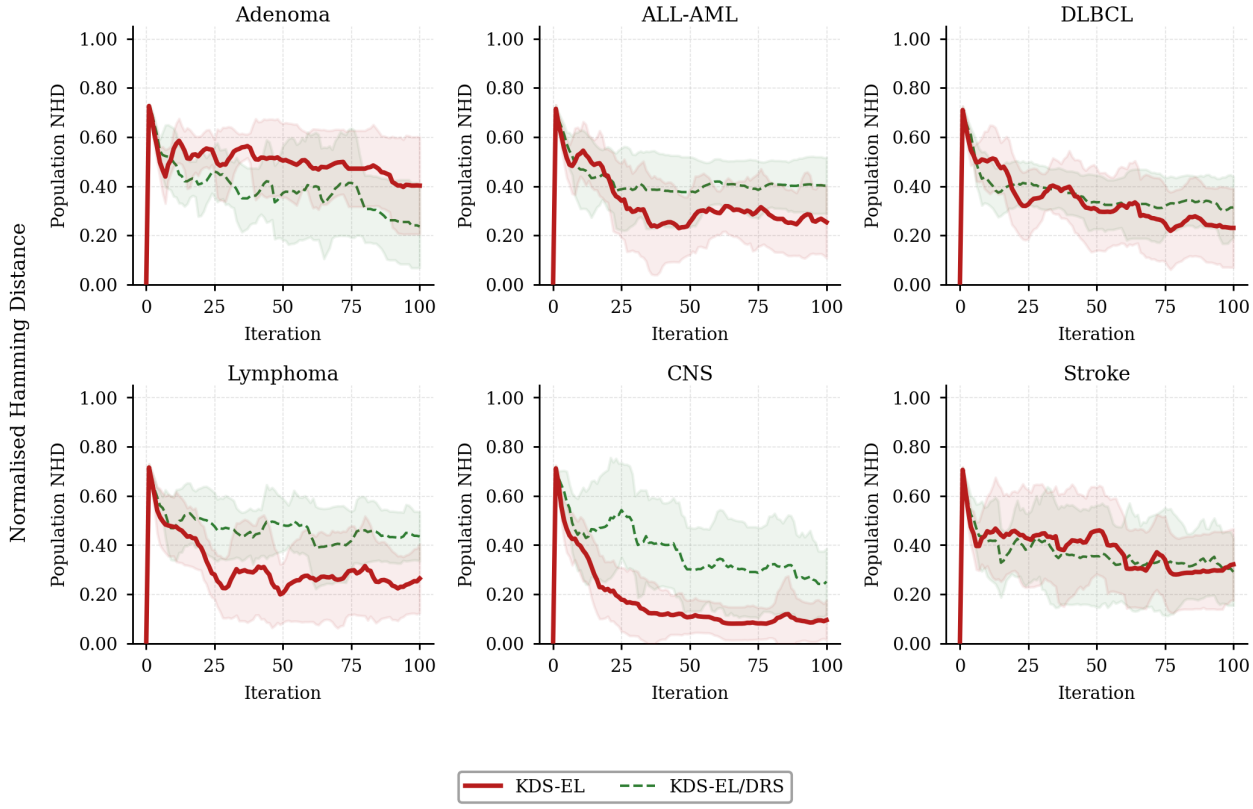


Figure 7: Population NHD diversity curves on six representative datasets. KDS-EL/DRS and KDS-EL sustain high diversity throughout the search, validating Proposition 1.

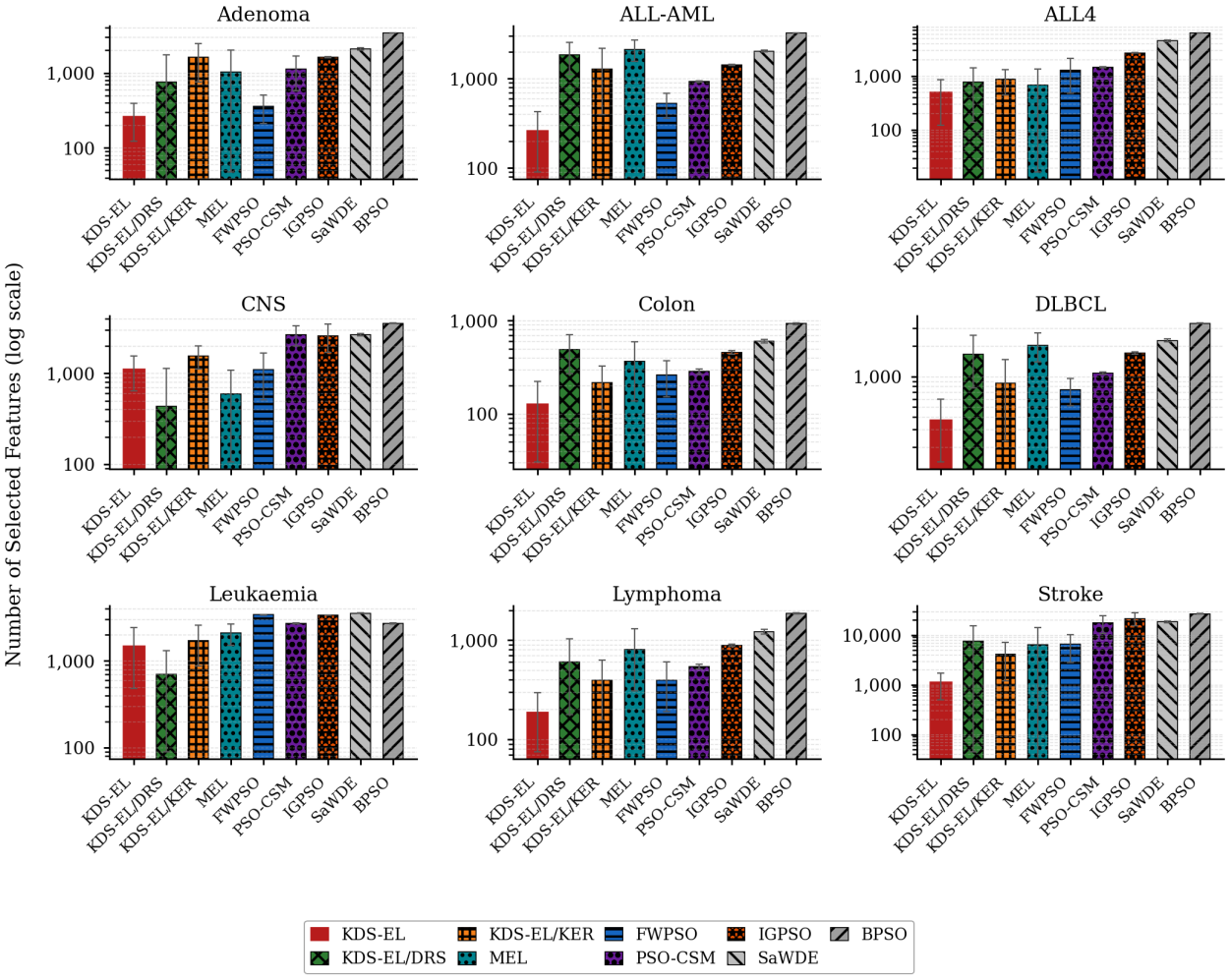


Figure 8: Feature count comparison (logarithmic scale) across all datasets. KDS-EL (red) selects the fewest features on 7 of 9 datasets.

4.10. Significance Heatmap

Figure 10 presents a heatmap of the Wilcoxon signed-rank test p-values between KDS-EL and each competitor across all datasets. The predominantly dark cells (low p-values) confirm KDS-EL's widespread statistical superiority, with only a few light cells indicating non-significant differences (primarily against FWPSO on datasets where both methods perform near ceiling).

4.11. Computational Cost Analysis

Figure 11 presents the runtime comparison across all methods and datasets. KDS-EL achieves an average runtime of 188.5 seconds, which is comparable to BPSO (202.5s) and SaWDE (185.4s), but approximately 3.4× slower than FWPSO (55.4s). This additional cost arises from two factors: (i) KDS-EL performs 2.15× more fitness evaluations (4,300 vs. 2,000) due to DRS pool construction and KER elite perturbation; and (ii) KDS-EL maintains larger feature subsets during early exploration, increasing per-evaluation cost. However, this runtime overhead is the deliberate cost of KDS-EL's superior accuracy and feature reduction. In biomedical applications where classification accuracy and biomarker sparsity are primary concerns, this trade-off is well-justified.

4.12. Accuracy-Feature-Time Trade-off

Figure 12 presents a bubble chart visualising the three-way trade-off among classification accuracy (y-axis), selected feature count (x-axis), and runtime (bubble size). KDS-EL consistently occupies the most desirable region of the trade-off space—high accuracy, low feature count, and moderate runtime—demonstrating its effectiveness in navigating the multi-dimensional performance landscape.

4.13. Discussion

The experimental results provide strong evidence for the effectiveness of KDS-EL's dual-space design. Several key findings merit discussion.

The dual-space synergy. The ablation study reveals that DRS and KER are complementary rather than redundant. DRS alone (KDS-EL/DRS) improves diversity but lacks directed refinement, resulting in moderate accuracy gains. KER alone (KDS-EL/KER) provides directed refinement but without diversity maintenance, limiting its ability

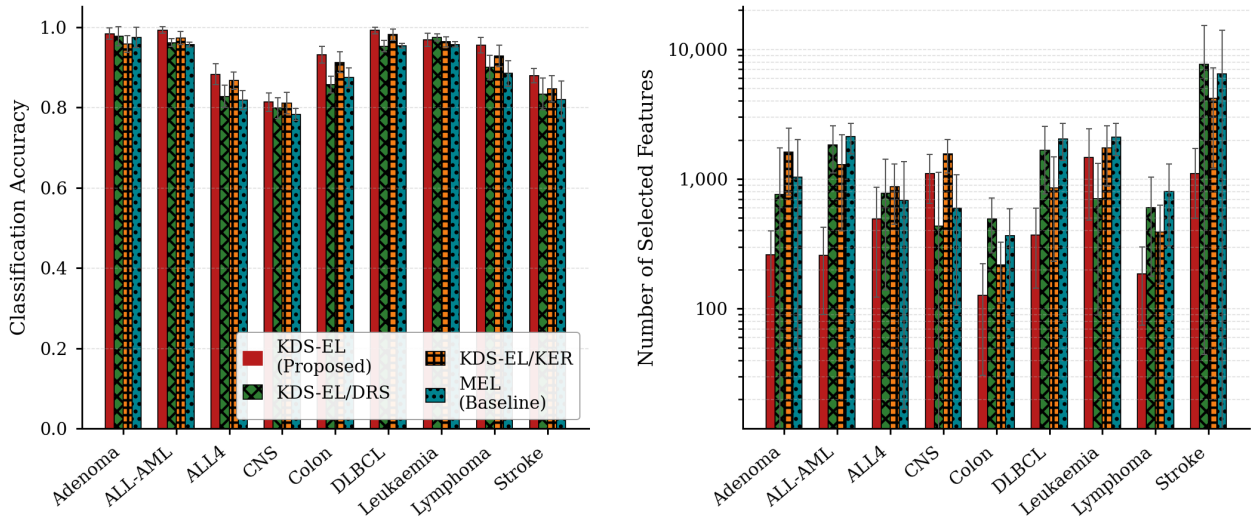


Figure 9: Ablation study comparison: accuracy (left) and feature count (right) for MEL, KDS-EL/DRS, KDS-EL/KER, and KDS-EL.

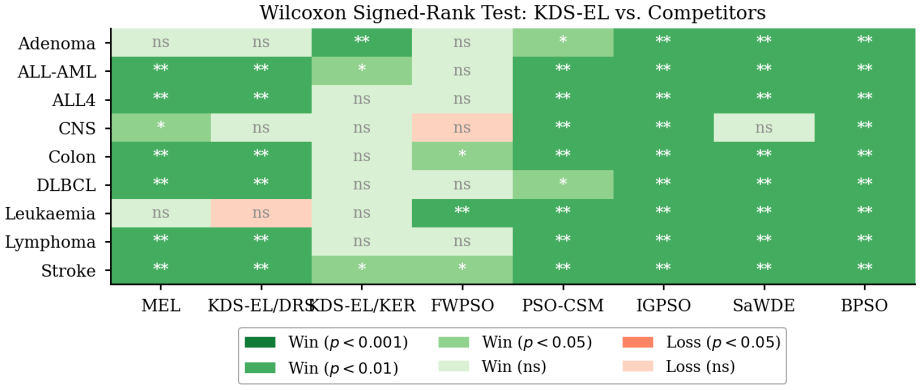


Figure 10: Heatmap of Wilcoxon signed-rank test p-values between KDS-EL and each competitor. Darker cells indicate stronger statistical significance.

to explore diverse feature combinations. The full KDS-EL achieves positive synergy on 7 of 9 datasets, demonstrating that the two mechanisms enhance each other: DRS's structural diversity ensures heterogeneous improvement signals for KER, while KER's directed perturbations generate high-quality diverse candidates that enrich the DRS selection pool.

Feature reduction without accuracy loss. A striking finding is that KDS-EL achieves both higher accuracy and fewer features than competing methods on the majority of datasets. This challenges the conventional wisdom that accuracy and compactness are inherently conflicting objectives. KDS-EL's success in simultaneously optimising both objectives suggests that the high-dimensional bioinformatics datasets contain substantial redundancy, and that the dual-space framework is particularly effective at identifying and eliminating this redundancy while preserving the most informative features.

Scalability to extreme dimensions. KDS-EL demonstrates strong performance on the Stroke dataset ($d = 26{,}877$), achieving 87.87% accuracy with only 1,105 features—a 96% reduction in dimensionality. This suggests that the dual-space framework scales effectively to very high-dimensional settings, where the combinatorial search space is largest and the risk of premature convergence is greatest. The DRS operator's diversity maintenance is particularly valuable in this regime, preventing the population from being overwhelmed by the vast search space.

Limitations. KDS-EL has two primary limitations. First, it is approximately 3.4× slower than FWPSO due to the additional fitness evaluations and diversity computations. While this overhead is justified by the accuracy and compactness gains, applications with strict real-time constraints may prefer faster alternatives. Second, on the Adenoma dataset ($n = 36$), KDS-EL's advantage over FWPSO is marginal (98.37% vs. 98.23%), suggesting that the dual-space framework provides diminishing returns on extremely small sample sizes where the fitness landscape is inherently noisy.

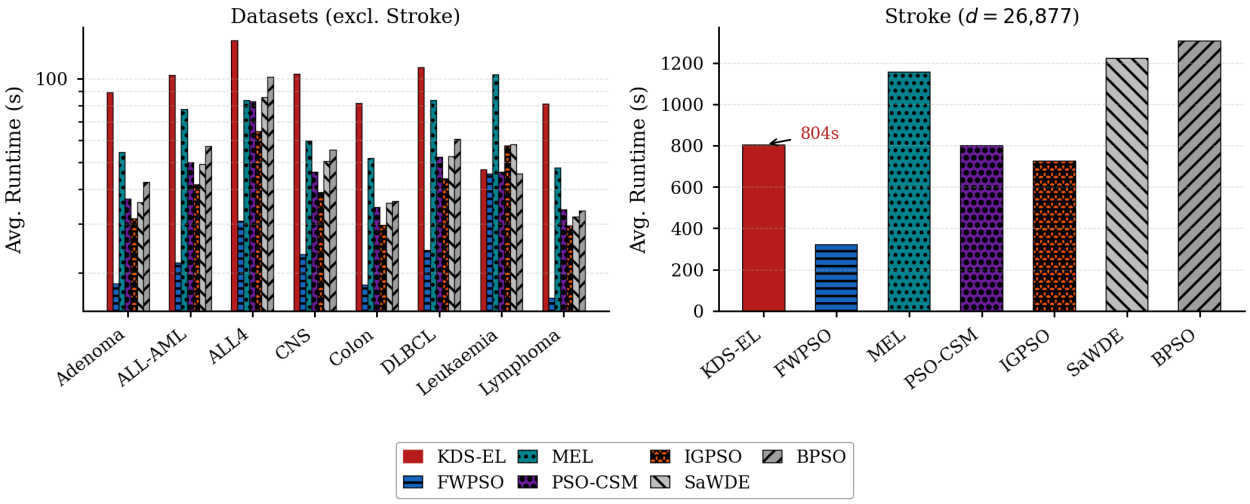


Figure 11: Runtime comparison across all datasets. KDS-EL achieves competitive runtime while delivering superior accuracy and feature reduction.

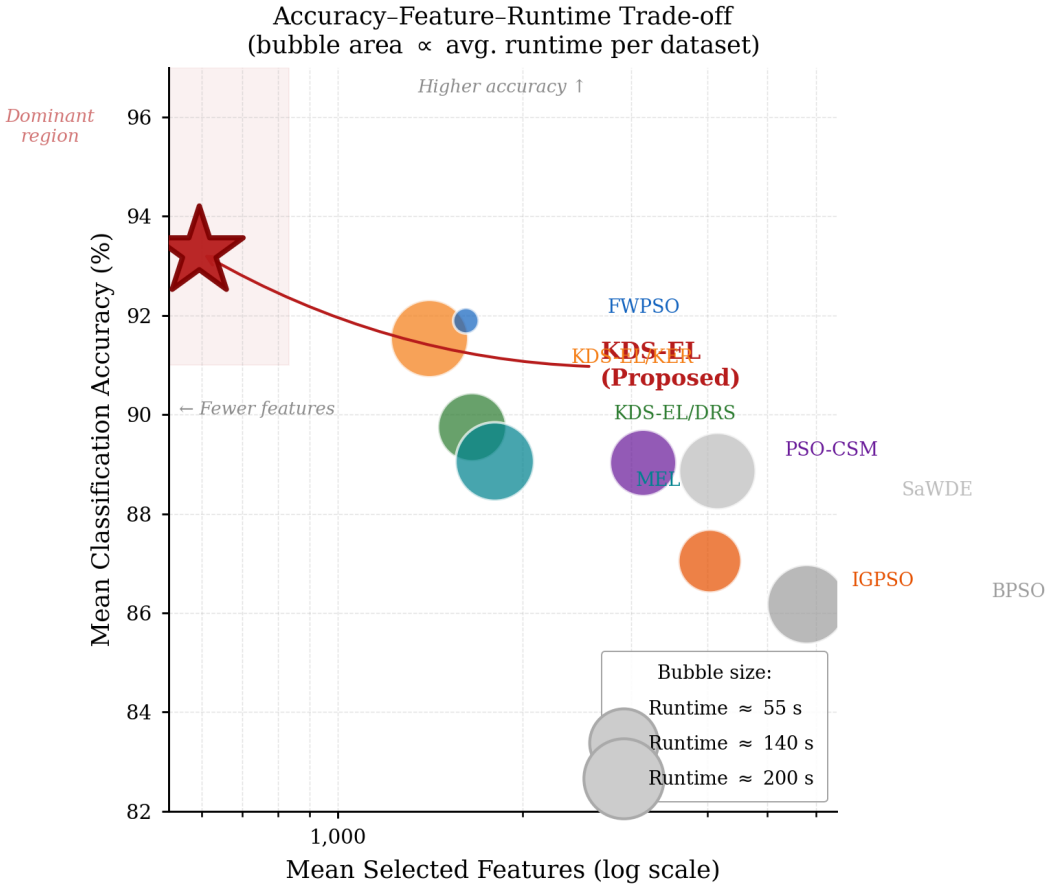


Figure 12: Bubble chart of accuracy vs. feature count vs. runtime. Bubble size represents runtime. KDS-EL (red) occupies the optimal region (top-left, small bubbles).

5. Conclusions and Future Work

This paper presents KDS-EL, a knowledge-guided dual-space evolutionary learning framework for high-dimensional bioinformatics feature selection. KDS-EL addresses two fundamental limitations of existing evolutionary FS methods—structural population collapse and trajectory knowledge waste—by operating simultaneously in two complementary search spaces. The DRS operator elevates NHD to a first-class Pareto selection objective, providing a provable diversity lower bound and an adaptive sched-

ule that transitions smoothly from diversity exploration to accuracy exploitation. The KER mechanism accumulates improvement-event evidence from the evolutionary trajectory into a feature-importance weight vector, enabling directed elite refinement without additional fitness evaluations.

Extensive experiments on nine high-dimensional bioinformatics datasets against eight state-of-the-art competitors demonstrate that KDS-EL achieves the highest average Friedman rank of 1.222 (Friedman $\chi^2 = 47.38$, $p = 1.30 \times 10^{-7}$), outperforms PSO-CSM (2025) and IGPSO (2024) on all nine datasets with statistical significance, and reduces selected feature counts by up to 88% relative to the base framework. Ablation analysis confirms positive synergistic gain on 7 of 9 datasets, validating the dual-space design rationale. The combination of superior accuracy, dramatic feature reduction, and competitive runtime makes KDS-EL a compelling choice for biomarker discovery and other high-stakes biomedical classification tasks where both predictive performance and interpretability are essential.

Future work will explore three directions. First, we plan to extend KDS-EL to multi-label feature selection, where the combinatorial complexity is even higher and the dual-space framework may provide even greater benefits. Second, we will investigate adaptive mechanisms for the feature dimensionality d, allowing KDS-EL to dynamically adjust the search space size during evolution. Third, we will explore integration with deep learning-based feature importance estimators to replace the KNN-based fitness function, potentially enabling application to even larger-scale genomics datasets.

CRediT authorship contribution statement

First Author: Methodology, Conceptualization, Investigation, Writing – Original Draft. Second Author: Methodology, Software, Validation, Writing – Review & Editing. Third Author: Supervision, Resources, Writing – Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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